Air Compressor



Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 7 labs

Covers these lab items: air compressor; Air Compressor (Small); Compressor

What it is used for

An air compressor takes in air and delivers it at increased pressure for compatible tools and equipment. Correct pressure settings, drainage, and safe hose handling help prevent injury and equipment failure.

Main parts

- Electric motor or engine drives the compressor mechanism.
- Pump compresses incoming air.
- Air receiver tank stores compressed air on tank-equipped models.
- Pressure gauge displays tank or outlet pressure.
- Pressure regulator adjusts delivery pressure for connected equipment.
- Safety relief valve releases excess pressure when required.
- Air outlet, hose, and couplings deliver compressed air to the tool.

Before you start

1. Read the compressor manual and confirm its pressure and air-flow ratings suit the task.
2. Allow only trained users to operate or service the compressor.
3. Inspect the tank, hoses, couplings, power cord, plug, gauges, and guards for damage.
4. Place the compressor on a stable, ventilated surface with sufficient cooling clearance.
5. Check the oil level if the model requires lubrication and verify the air filter is clear.
6. Ensure the drain valve is closed before starting and the safety relief valve is unobstructed.

How to use

1. Switch off the compressor and release stored pressure before connecting or changing accessories.
2. Connect the hose securely using compatible couplings and check the tool's pressure requirement.
3. Set the regulator to the pressure specified for the connected tool.
4. Connect the compressor to the correct power supply or prepare its approved engine start procedure.
5. Start the compressor and allow the receiver tank to build pressure within its rated limit.
6. Check the gauges and listen for unusual noise, vibration, or air leaks.
7. Operate the connected tool using controlled movements and the recommended air pressure.
8. Stop the compressor if pressure behaves abnormally, the motor overheats, or a hose is damaged.

9. Switch off and isolate power before maintenance, and release pressure according to the manual.
10. Drain moisture from the receiver tank as instructed and store hoses neatly after use.

⚠ Safety

- Never exceed the rated pressure of the tank, hose, regulator, or connected tool.
- Never point compressed air at people, skin, or clothing.
- Do not use compressed air to clean skin or blow dust from the body.
- Never repair, weld, or modify a pressurised air receiver tank.
- Keep hands away from moving belts, pulleys, and hot compressor surfaces.
- Depressurise the system before disconnecting hoses or clearing blockages.

Care & storage

- Drain condensate at the interval specified by the manufacturer to reduce corrosion.
- Inspect hoses, couplings, gauges, guards, and power cables regularly.
- Clean or replace the air filter and service lubrication systems as directed.
- Arrange authorised inspection if the tank corrodes, the relief valve fails, or pressure leaks persist.

Common mistakes to avoid

- Setting the regulator above the connected tool's required pressure.
- Disconnecting a hose while the system remains pressurised.
- Ignoring water accumulation, air leaks, or damaged hoses.
- Using compressed air to clean clothing or skin.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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